

ARACHIS ONLINE TRAINING

GUMMY SWEETS PRODUCTION MODULE

Affordable Commercial Gummies Suitable for Zimbabwe

INTRODUCTION

Welcome to the **Arachis Online Training Gummy Production Module**.

In this lesson you will learn:

- * What gummies are
- * Ingredients used in gummy production
- * The purpose of each ingredient
- * Step-by-step production procedure
- * Common mistakes
- * Cost reduction methods
- * Packaging ideas
- * Business tips for Zimbabwe

This module is designed for:

- * beginners
- * small business owners
- * students starting with low capital

The formula below is:

- * affordable
- * practical
- * suitable for small-scale production in Zimbabwe

WHAT ARE GUMMIES?

Gummies are:

soft chewy sweets made using a gelling agent.

The gelling agent creates the rubbery chewy texture.

There are two common types:

Type	Main Gelling Agent
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Gelatin gummies	Gelatin
Vegan gummies	Pectin

In this module we use:

GELATIN

because it is:

- * easier to work with
- * cheaper
- * beginner friendly
- * gives excellent chewiness

UNDERSTANDING THE INGREDIENTS

1. SUGAR

Function:

- * Sweetness
- * Body
- * Preservation

Sugar is the main solid ingredient.

Too little sugar:

- * weak taste
- * poor texture

Too much sugar:

- * gummies become too hard

2. GLUCOSE SYRUP

Function:

- * Prevents crystallization
- * Improves chewiness
- * Makes gummies smoother

Without glucose:

- * gummies may become grainy
- * sugar crystals may form

Important:

Glucose syrup is one of the ingredients that gives commercial gummies their chewy texture.

3. GELATIN

Function:

- * Creates the gummy structure
- * Gives elasticity
- * Gives chewiness

Gelatin is the heart of gummy production.

Too little gelatin:

- * gummies become weak and watery

Too much gelatin:

- * gummies become rubber-like and too hard

4. WATER

Function:

- * Dissolves ingredients
- * Helps hydration

Water must be controlled carefully.

Too much water:

- * sticky gummies
- * poor shelf life

5. CITRIC ACID

Function:

- * Sour taste
- * Flavor enhancement

Citric acid makes fruit flavors taste stronger and fresher.

Too much:

- * overly sour gummies
- * may weaken gelatin

6. FLAVORS

Function:

- * Gives identity to the product

Popular Zimbabwean flavors:

- * Orange
- * Raspberry
- * Strawberry
- * Banana
- * Cream Soda
- * Pineapple

7. FOOD COLOR

Function:

- * Improves appearance
- * Makes products attractive

Bright colors increase customer appeal especially for children.

FORMULA — 1KG GUMMY BATCH

Ingredient	Amount	
-----	-----	
Sugar	350g	
Glucose Syrup	200g	
Water	180ml	
Gelatin	80g	
Citric Acid	5g	
Flavor	5–10ml	

| Food Color | Small amount |

EQUIPMENT REQUIRED

Equipment	Purpose	
-----	-----	
Pot	Cooking	
Stove	Heating	
Measuring scale	Accuracy	
Spoon	Mixing	
Silicone molds	Shaping	
Thermometer	Temperature control	

PRODUCTION PROCEDURE

STEP 1 — PREPARE THE GELATIN

This stage is called:

BLOOMING

Procedure

Take:

- * 80g gelatin
- * 100ml cold water

Mix together.

Leave for:

- * 10–15 minutes

What Happens During Blooming?

The gelatin absorbs water and swells.

This prepares it for smooth melting later.

Why Blooming Is Important

Without blooming:

- * lumps form
- * poor texture develops
- * gummies become uneven

STEP 2 — PREPARE THE SUGAR SYRUP

In another pot add:

- * 350g sugar
- * 200g glucose syrup
- * 80ml water

Heating Procedure

Heat slowly while stirring.

Do NOT use very high heat immediately.

Continue heating until:

- * sugar dissolves completely
- * mixture thickens slightly

Ideal Temperature

Commercial gummy cooking is usually around:

110°C to 115°C

Why Temperature Matters

If temperature is too low:

- * gummies become too soft

If temperature is too high:

- * gummies become too hard
- * sugar may burn

STEP 3 — ADD THE GELATIN

Reduce heat.

Add bloomed gelatin slowly.

Stir gently until fully dissolved.

VERY IMPORTANT WARNING

Do NOT aggressively boil gelatin.

High heat destroys gelatin strength.

Destroyed gelatin causes:

- * weak gummies
- * sticky texture
- * poor setting

STEP 4 — ADD FLAVOR, COLOR & CITRIC ACID

Now add:

- * flavor
- * food color
- * citric acid

Mix thoroughly.

IMPORTANT NOTE ABOUT CITRIC ACID

Citric acid should usually be added:

- * near the end

because too much heat weakens sourness.

STEP 5 — POUR INTO MOLDS

Pour mixture quickly into molds.

Why quickly?

Because gummies begin setting as they cool.

STEP 6 — SETTING

Allow gummies to set:

- * 4–6 hours

OR

- * overnight

Best results:

- * room temperature

- * dust-free environment

STEP 7 — DRYING

After removing from molds:

- * dry gummies for 12–24 hours

WHY DRYING IS IMPORTANT

Drying:

- * improves chewiness

- * reduces stickiness

- * increases shelf life

STEP 8 — DUSTING

Lightly coat gummies with:

- * cornstarch

OR

- * icing sugar

This prevents sticking during packaging.

COST REDUCTION METHODS

1. Use Less Expensive Flavors

Simple flavors:

- * orange
- * banana
- * raspberry

are cheaper than imported premium flavors.

2. Buy Gelatin in Bulk

Small sachets are expensive.

Buying larger quantities reduces cost greatly.

3. Repackage into Small Units

This is very important in Zimbabwe.

Instead of selling:

- * 1kg packs

sell:

- * \$0.20 packs
- * \$0.50 packs
- * \$1 packs

This improves:

- * affordability
- * customer reach
- * profit margins

COMMON PROBLEMS & SOLUTIONS

Problem	Cause	Solution
-----	-----	-----
Gummies too soft	Too much water	Reduce water
Gummies too hard	Too much gelatin	Reduce gelatin
Sticky gummies	Poor drying	Dry longer
Grainy gummies	Poor glucose level	Increase glucose
Weak flavor	Too little flavor	Increase flavor
Gummies melt easily	Weak gelatin structure	Improve cooking

HYGIENE & SAFETY

Always:

- * wash equipment
- * use clean water
- * sanitize molds
- * cover products properly

Poor hygiene causes:

- * contamination
- * short shelf life
- * customer complaints

PACKAGING IDEAS

Recommended packaging:

- * transparent pouches
- * stand-up pouches
- * heat-sealed sachets

IMPORTANT BUSINESS LESSON

Customers often judge quality using:

- * packaging
- * color
- * branding

before tasting the product.

Good packaging allows you to:

- * charge higher prices
- * attract better customers
- * compete with bigger brands

ESTIMATED BUSINESS COSTING (ZIMBABWE)

Approximate production cost:

- * around \$4–\$7 per kg

Possible selling value after repacking:

- * \$10–\$20+ per kg equivalent

Most profit comes from:

SMALL PACKAGING

FINAL ADVICE TO STUDENTS

Start simple.

Do not wait for perfect equipment.

You can begin with:

- * a pot
- * stove
- * molds
- * proper measurements

Focus on:

- * consistency
- * flavor
- * cleanliness
- * packaging

As sales grow:

- * improve branding
- * introduce new flavors
- * scale production

END OF MODULE

ARACHIS ONLINE TRAINING

“Practical Skills for Real Business”